Stochastic Recovery from Information Loss and Equality of Channel Capacity

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Abstract

For finite channels, a Bayesian reverse kernel converts mutual-information loss under a specified full-support probe into explicit weighted relative-entropy and total-variation recovery bounds. Zero loss is equivalent to a common stochastic recovery of every input row. Equal channel capacities therefore imply recovery when the coarse channel has a full-support optimizer, but existence of a full-support optimizer for the original channel alone is insufficient, even for strictly positive channels. For a specified three-input, four-output example with equal capacities before and after output merging, we prove that the sharp worst-case per-input recovery error is one quarter. The same fixed example has arbitrarily small information loss under a sequence of full-support probes, showing why a quantitative probe-mass condition matters. All conclusions concern explicit population probability models; finite-sample coverage and physical correspondence require separate premises.

1 Channels, probes and recovery

Let $\mathcal{X}, \mathcal{Y}, \mathcal{V}$ be nonempty finite alphabets. Write probability distributions as row vectors. A channel W from $\mathcal{X}$ to $\mathcal{Y}$ and a stochastic observation map K from $\mathcal{Y}$ to $\mathcal{V}$ give the coarse channel

$$Q = WK, \qquad Q(v \mid x) = \sum_{y} W(y \mid x) K(v \mid y).$$

A recovery channel is a single stochastic matrix L from $\mathcal{V}$ to $\mathcal{Y}$. It may depend on the specified channels and probe, but it may not depend on the unknown input x. Its recovered channel is QL. Exact recovery means $QL = W$ on every input row.

Throughout, relative entropy and mutual information use base-2 logarithms. Our total-variation convention is

$$\mathrm{TV}(a, b) = \frac{1}{2} \sum_{y} |a_y - b_y|.$$

For an input probe p, the information loss is

$$\delta_p = I_p(X;Y) - I_p(X;V) = I_p(X;Y \mid V) \geq 0, \qquad X \longrightarrow Y \longrightarrow V. \tag{1.1}$$

The identity follows from the mutual-information chain rule and the Markov property. These are the usual data-processing and sufficiency relations [1, 2]. No physical time, frequency, energy or apparatus interpretation enters these finite probability models.

Physical use of a channel requires a further correspondence to the available inputs and recorded outputs.

We distinguish a probe-specific average from the uniform recovery error

$$\mathfrak{r}_\infty(W,K) = \min_{L \text{ stochastic}} \max_{x\in\mathcal{X}} \mathrm{TV}(W_x,(QL)_x). \tag{1.2}$$

The minimum exists: stochastic matrices L form a compact product of simplexes and the objective is continuous. This definition permits all stochastic recovery maps, including ones that do not respect the fibres of a deterministic observation map.

2 An explicit quantitative recovery map

Fix a full-support probe $p_x > 0$ for every x. Put $\mu = pW$ and $\nu = \mu K = pQ$. Define the Bayesian reverse channel

$$L_p(y \mid v) = \frac{\mu_y K(v \mid y)}{\nu_v} \quad \text{when } \nu_v > 0. \tag{2.1}$$

For $\nu_v = 0$, choose any probability row. Full support of p then implies $Q(v \mid x) = 0$ for every x, so that arbitrary row has no effect on QL_p. Let $R_p = QL_p$ and $e_x = \mathrm{TV}(W_x,(R_p)_x)$.

Theorem 2.1 (Information loss controls recovery)**.** *Under the preceding finite-alphabet assumptions,*

$$\sum_x p_x D_2(W_x\|(R_p)_x) \le \delta_p. \tag{2.2}$$

If K is deterministic, equality holds in (2.2)*. For an arbitrary stochastic K,*

$$\sum_x p_x e_x^2 \le \frac{\ln 2}{2}\delta_p, \qquad \sum_x p_x e_x \le \sqrt{\frac{\ln 2}{2}\delta_p}, \tag{2.3}$$

$$e_x \le \min\left\{1, \sqrt{\frac{\ln 2}{2p_x}\delta_p}\right\}, \qquad \mathfrak{r}_\infty(W,K) \le \min\left\{1, \sqrt{\frac{\ln 2}{2p_{\min}}\delta_p}\right\}, \tag{2.4}$$

where $p_{\min} = \min_x p_x$.

Proof. On triples (x,y,v) form the two probability laws

$$J(x,y,v) = p_x W(y \mid x) K(v \mid y), \qquad M(x,y,v) = p_x Q(v \mid x) L_p(y \mid v).$$

The second law is obtained from the (X,V) marginal of the first by resampling Y from its V-conditional marginal. Consequently

$$D_2(J\|M) = I_J(X;Y \mid V) = \delta_p.$$

There is no support difficulty: $J(x,y,v) > 0$ implies $Q(v \mid x) > 0$, $\mu_y > 0$, $\nu_v > 0$ and $L_p(y \mid v) > 0$. Marginalizing over v gives the laws $p_x W(y \mid x)$ and $p_x(R_p)(y \mid x)$. The log-sum inequality, applied separately at each (x,y), therefore gives

$$\delta_p = D_2(J\|M) \ge D_2(J_{XY}\|M_{XY}) = \sum_x p_x D_2(W_x\|(R_p)_x).$$

Here $J_{XY}(x,y) = p_x W(y \mid x)$ and $M_{XY}(x,y) = p_x(R_p)(y \mid x)$ are joint laws.

If $K(v \mid y) = \mathbf{1}\{v = f(y)\}$, both J and M are supported on $v = f(y)$. The marginalization is then one-to-one on their common support and loses no relative entropy. This proves the equality claim.

Pinsker's inequality in the stated convention is

$$\mathrm{TV}(a, b)^2 \le \frac{\ln 2}{2} D_2(a\|b).$$

Apply it row by row and sum to obtain the first bound in (2.3). Cauchy–Schwarz gives the second because $\sum_x p_x = 1$. Each nonnegative term $p_x e_x^2$ is at most the weighted sum, giving the first bound in (2.4). The chosen L_p is an admissible recovery map, so taking its maximum row error also bounds (1.2). □

For completeness, the constant in Pinsker's inequality can be checked without a convention-dependent appeal. Let $A = \{y : a_y \ge b_y\}$, $u = a(A)$ and $v = b(A)$; then $u - v = \mathrm{TV}(a, b)$. Log-sum gives $D_{\mathrm{nat}}(a\|b) \ge d_{\mathrm{nat}}(u\|v)$. For fixed $0 < v < 1$, the latter function of u is zero with zero derivative at $u = v$ and has second derivative $1/[u(1-u)] \ge 4$. Integrating twice gives $d_{\mathrm{nat}}(u\|v) \ge 2(u - v)^2$; endpoints follow by limits or infinite divergence. Finally $D_{\mathrm{nat}} = (\ln 2) D_2$. Sason and Verdú state the same inequality using twice our total-variation normalization [3, Eq. (1), Eqs. (55)–(58)].

The inequalities above give explicit sufficient bounds, not a sharp universal recovery modulus. The Bayesian reverse map need not minimize the maximum row error. A sharp minimax calculation for the example below illustrates that distinction.

3 Exact equality and full support

Theorem 3.1 (One full-support probe suffices for exact recovery)**.** *For finite W, K, the following statements are equivalent:*

1. *For some full-support input law p, $I_p(X;Y) = I_p(X;V)$.*
2. *There is a stochastic recovery matrix L such that $WKL = W$.*
3. *For every input law q, $I_q(X;Y) = I_q(X;V)$.*

When the first condition holds, (2.1) *is a recovery map. Moreover the actual joint law obeys $X \longrightarrow V \longrightarrow Y$ under each full-support probe.*

Proof. If the first condition holds, (2.2) is a sum of nonnegative terms with strictly positive weights and upper bound zero. Each divergence vanishes, so $W_x = (WKL_p)_x$ for every x. This proves the second condition. If the second holds, data processing applied successively to W, K and L gives, for every q,

$$I_q(X;Y) \ge I_q(X;V) \ge I_q(X;Y'),$$

where Y' is the recovered output. Since Y' has conditional law W, the first and last terms coincide, proving the third condition. The third implies the first by choosing any full-support law. Finally (1.1) makes conditional mutual information zero. Its conditional-divergence expression implies conditional independence of X and Y given V. □

This is the finite-experiment sufficiency criterion expressed with a single probe that charges every input. The conventional formulation quantifies over all priors [2, Theorem 3.1]. If a probe has zero mass on an input, the argument establishes recovery only on its support. It cannot constrain an untested row.

Finite full support means positive mass on every input symbol. On an uncountable alphabet, a distribution can have full topological support while assigning zero mass to an individual input. Changing a channel only at that input leaves its mutual information under the probe unchanged. Pointwise recovery in such a setting requires additional regularity or an explicitly almost-everywhere conclusion; it is not an automatic extension of Theorem 3.1.

4 What equality of capacities implies

Let $\mathcal{F}$ be the same nonempty family of admissible input laws for both channels, including any input-cost restriction. Define

$$C_{\mathcal{F}}(W) = \sup_{p \in \mathcal{F}} I_p(X;Y), \qquad C_{\mathcal{F}}(Q) = \sup_{p \in \mathcal{F}} I_p(X;V), \qquad g = C_{\mathcal{F}}(W) - C_{\mathcal{F}}(Q) \geq 0.$$

No change of admissible family is permitted in this comparison.

Corollary 4.1 (A full-support coarse-channel optimizer)**.** *Suppose $p \in \mathcal{F}$ has full support and $I_p(X;V) \geq C_{\mathcal{F}}(Q) - \eta$ for some $\eta \geq 0$. Then its reverse channel satisfies all bounds of Theorem 2.1 with δ_p replaced by $g + \eta$. In particular, equal capacities and a full-support maximizing input for the coarse channel imply exact recovery of every row of W.*

Proof. The definition of the two suprema gives

$$0 \leq \delta_p = I_p(X;Y) - I_p(X;V) \leq C_{\mathcal{F}}(W) - C_{\mathcal{F}}(Q) + \eta = g + \eta.$$

Apply Theorem 2.1. If $g = \eta = 0$, apply Theorem 3.1. □

Thus full support of a common capacity maximizer suffices. A related sufficient condition concerns uniqueness: if the coarse capacity is attained, the original channel has a unique capacity-maximizing law, and that law has full support, equality of capacities also implies recovery. Indeed, every coarse-channel maximizer must then maximize the original channel by the inequalities

$$C_{\mathcal{F}}(Q) = I_p(X;V) \leq I_p(X;Y) \leq C_{\mathcal{F}}(W).$$

Uniqueness makes it the stated full-support law. For finite channels with all input distributions allowed, both capacities are attained by continuity on the input simplex. An arbitrary nonclosed family $\mathcal{F}$ needs its own attainment assumption.

Example 4.2 (Full support for the original optimizer is insufficient)**.** Consider the four-input channel

$$\widetilde{W} = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \end{pmatrix}, \qquad K = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}.$$

All input distributions are admissible. Every row of $\widetilde{W}$ has entropy one bit. Therefore $I_p(X;Y) \leq 2 - 1 = 1$. The uniform input on all four symbols has full support and produces uniform output, attaining capacity one. The coarse channel has rows $(1,0)$, $(0,1)$, $(1/2,1/2)$, $(1/2,1/2)$ and also has capacity one, attained by equal use of the first two inputs.

Nevertheless its third and fourth rows coincide whereas those of $\widetilde{W}$ differ. Applying any common L to two equal rows keeps them equal, so recovery is impossible. At the full-support uniform input, coarse mutual information is only $1/2$ bit. That input maximizes the original channel but does not maximize the coarse channel. In fact

$$I_p(X;V) = h_2\left(p_1 + \frac{p_3+p_4}{2}\right) - (p_3+p_4)$$

attains one only when $p_3 = p_4 = 0$ and $p_1 = p_2 = 1/2$.

The same distinction persists with strictly positive channel entries. Let U be the four-input channel every row of which is uniform on four outputs, and set $\widetilde{W}^{(\rho)} = (1-\rho)\widetilde{W} + \rho U$ for $0 < \rho < 1$. Each row has two entries $(2-\rho)/4$ and two entries $\rho/4$, hence entropy $1 + h_2(\rho/2)$. The uniform input again attains

$$C(\widetilde{W}^{(\rho)}) = 1 - h_2(\rho/2).$$

The first two coarse rows form a binary symmetric channel with crossover probability $\rho/2$, whose equal input attains that same value. Data processing supplies the matching upper bound. The third and fourth coarse rows still coincide and their original rows still differ. Thus mere existence of a full-support optimizer for the original channel remains insufficient even in this positive class.

5 A sharp three-input recovery error

The three-input example in the companion capacity study [4, Proposition 6] is

$$W = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}, \qquad Q = WK = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \tag{5.1}$$

with the preceding merger K. Both capacities are one bit: all original row entropies equal one, and equal use of the first two rows attains the uniform four-output distribution and a noiseless binary coarse output. The original capacity-maximizing law is uniquely $(1/2, 1/2, 0)$, since a uniform original output forces the third input mass to vanish. There is no full-support capacity optimizer for either channel.

Lemma 5.1 (A preserved convex relation obstructs recovery)**.** *If $Q_{x_0} = \sum_i \lambda_i Q_{x_i}$, where $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$, then every recovery map has row errors $e_x = \mathrm{TV}(W_x, (QL)_x)$ obeying*

$$\mathrm{TV}\left(W_{x_0}, \sum_i \lambda_i W_{x_i}\right) \leq e_{x_0} + \sum_i \lambda_i e_{x_i}.$$

Consequently $\mathfrak{r}_\infty(W,K)$ is at least half the total-variation distance on the left.

Proof. Linearity gives $(QL)_{x_0} = \sum_i \lambda_i (QL)_{x_i}$. Insert this distribution between the two laws on the left, apply the triangle inequality, and then use convexity of total variation in the second comparison. Each error is at most their maximum. □

Theorem 5.2 (Exact minimax error and anchor tradeoff)**.** *For* (5.1)*,*

$$\mathfrak{r}_\infty(W,K) = \frac{1}{4}. \tag{5.2}$$

More precisely, for any $0 \leq a \leq 1/2$, requiring the first two row errors to be at most a forces the third error to be at least $1/2 - a$, and that bound is attainable.

Proof. Here $Q_3 = (Q_1 + Q_2)/2$, but

$$\mathrm{TV}\left(W_3, \frac{W_1 + W_2}{2}\right) = \mathrm{TV}((1/2, 0, 1/2, 0), (1/4, 1/4, 1/4, 1/4)) = \frac{1}{2}.$$

Lemma 5.1 gives $1/2 \le e_3 + (e_1 + e_2)/2$. If all three errors are at most t, then $t \ge 1/4$. If $e_1, e_2 \le a$, it gives $e_3 \ge 1/2 - a$.

For attainability, take

$$L_a = \begin{pmatrix} 1/2 + a & 1/2 - a & 0 & 0 \\ 0 & 0 & 1/2 + a & 1/2 - a \end{pmatrix}.$$

It is stochastic for the stated range. Direct subtraction gives

$$\left(\mathrm{TV}(W_i, (QL_a)_i)\right)_{i=1}^{3} = (a, a, 1/2 - a).$$

At $a = 1/4$, all three errors equal $1/4$, attaining the universal lower bound. This proves both statements. □

The value $1/2$ obtained by reconstructing the first two rows exactly is therefore an anchored recovery error, not the optimum in (1.2). The minimax map deliberately allows an error of $1/4$ on each of those two rows to reduce the error on the third.

6 Why a quantitative probe-mass condition is necessary

Keep the channel and merger in (5.1) fixed. For $0 < \varepsilon < 1$, use the full-support probe

$$p_\varepsilon = ((1 - \varepsilon)/2, (1 - \varepsilon)/2, \varepsilon).$$

The coarse output is uniform, while the original output is

$$\mu_\varepsilon = ((1 + \varepsilon)/4, (1 - \varepsilon)/4, (1 + \varepsilon)/4, (1 - \varepsilon)/4).$$

It follows directly that

$$\begin{aligned} I_{p_\varepsilon}(X; Y) &= h_2((1 + \varepsilon)/2), \\ I_{p_\varepsilon}(X; V) &= 1 - \varepsilon, \\ \delta_{p_\varepsilon} &= h_2((1 + \varepsilon)/2) - 1 + \varepsilon. \end{aligned} \tag{6.1}$$

Strict concavity of binary entropy between $1/2$ and 1 gives $h_2((1 + \varepsilon)/2) > 1 - \varepsilon$, while its maximum is one. Thus $0 < \delta_{p_\varepsilon} < \varepsilon$ and the loss tends to zero with ε.

The experiment has not changed, so Theorem 5.2 still gives $\mathfrak{r}_\infty = 1/4$. Therefore no bound depending only on δ_p, and tending to zero with it, can control worst-case row recovery error uniformly over all full-support probes, even for these fixed alphabet sizes. Full support alone is insufficient; a uniform guarantee needs further restrictions. A quantitative lower bound $p_x \ge \alpha > 0$ is one sufficient condition, as (2.4) shows. This example does not assert optimality of that particular bound or its dependence on α.

In this family the Bayesian reverse map is

$$L_{p_\varepsilon} = \begin{pmatrix} (1 + \varepsilon)/2 & (1 - \varepsilon)/2 & 0 & 0 \\ 0 & 0 & (1 + \varepsilon)/2 & (1 - \varepsilon)/2 \end{pmatrix},$$

with row errors $(\varepsilon/2, \varepsilon/2, (1 - \varepsilon)/2)$. For the uniform probe, $\varepsilon = 1/3$, its maximum error is $1/3$, larger than the minimax value $1/4$. At $\varepsilon = 1/2$ it equals the minimax map. Bayesian recovery, probe weighting and minimax recovery are different optimization choices.

7 Interpretation and verification boundary

These results separate three questions. A channel representation defines a mathematical optimization problem. Retention of its full experiment concerns common stochastic recovery of every input row. A physical interpretation additionally concerns which preparations, observation maps and recovery operations the apparatus realizes. Equality of capacities alone settles neither the latter two questions. Exact information equality under a finite full-support probe does settle stochastic recoverability, without supplying an underlying physical ontology.

The quantitative bounds use population channel laws and a specified probe. An estimated information loss is not automatically an upper confidence bound on δ_p. Sampling dependence, estimation bias, calibration uncertainty, optimization error and selection of the probe require their own justified treatment before the bounds can be used as experimental guarantees.

The supplied verification program checks the rational channel and recovery matrices, the attained $1/4$ errors, the anchor tradeoff, the positive four-input counterexample, a recoverable positive control, unused output labels, and the rare-probe formulas. A nontrivial stochastic observation map also checks the joint-law construction. Rational identities are exact. Evaluations involving logarithms use 80-digit arithmetic as finite numerical checks; they are not interval enclosures. The universal statements follow from the displayed proofs, not from the finite program. The work uses shared source material and the same model lineage as the companion studies. No independent replication, empirical validation or literature-priority claim is made.

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